Kasra Siavashpour

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EDUCATION

Sharif University of Technology (SUT), Tehran, Iran — Nov 2020–Jul 2026
Bachelor of Science, Computer Science, Department of Mathematical Sciences

Allameh Tabatabaei High School (selective, non-governmental), Tehran, Iran — 2016–2020
High School Diploma, Mathematics and Physics

RESEARCH INTERESTS

- Distributed & Decentralized Systems
- IoT & Cyber-Physical Systems
- Multi-Agent Systems
- AI-Driven Network Automation

PUBLICATIONS

Submitted

- A. Vaezi, S. Niknejad, **K. Siavashpour**, A. Dadras, *"FAIR: An Envy-Free Multi-Agent Model for Traffic Systems,"* submitted to the **International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)**.

In Preparation

- A. Vaezi, A. Daneshgar, A. Dadras, **K. Siavashpour** (author order not final), *"The Loop of the Rings: A Fully Decentralized Reliable Cooperative System,"* manuscript in preparation for submission to the **IEEE Symposium on Foundations of Computer Science (FOCS) 2027**. [arXiv preprint]
- A. Siavashpour, **K. Siavashpour**, M. Azadegan, B. Safaei, H. Peyvandi, M. Imran, *"A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems,"* to be submitted to the **IEEE Open Journal of the Communications Society (OJ-COMS)**.
- S. Azarnoush, D. Babaei, M. Ghodsi, A. Hashemi, A. Khanari, M. H. Mehdikhani, P. Mohammadian, L. Motahari, **K. Siavashpour**, A. Vaezi (alphabetical order), *"Operational Resilience of Blockchain Remote Procedure Call Services: Exposure, Authority Boundaries, and the Transition to External Signing,"* to be submitted to **Blockchain: Research and Applications** (Elsevier).

RESEARCH EXPERIENCE

Research Assistant, **Institute for Research in Fundamental Sciences (IPM)**
& **Sharif University of Technology**, Tehran, Iran — May 2026–present
Supervisor: Dr. Arash Vaezi
Project: "FAIR: An Envy-Free Multi-Agent Model for Traffic Systems"

- Strengthened the fairness theorem and contributed to reformulating two preliminary lemmas.
- Built an event-driven simulator to evaluate FAIR on three real-world road networks (up to 93K vehicles).
- Introduced traffic-model refinements adopted in the final model and identified a dimensional inconsistency.
- FAIR achieved 2.5–10× faster network clearance and more predictable travel times under congestion.

Research Assistant, **Institute for Research in Fundamental Sciences (IPM)**
& **Sharif University of Technology**, Tehran, Iran — Aug 2024–present
Supervisors: Dr. Arash Vaezi, Dr. Amir Daneshgar
Project: "The Loop of the Rings: A Fully Decentralized Reliable Cooperative System"

- Analyzed LoR's randomized ring structure to evaluate protocol reliability under adversarial trader behavior.
- Investigated reliability and user satisfaction in networks with at least 60% honest traders.
- Designed and ran 421 simulations of 500 traders to validate theoretical reliability under adversarial conditions.
- Implemented the core protocol in Go and evaluated its behavior under worst-case adversarial conditions.

Research Assistant, **Sharif University of Technology**, Tehran, Iran — Aug 2024–present
Supervisors: Dr. Hossein Peyvandi, Dr. Masoumeh Azadegan (Khatam University),
Dr. Bardia Safaei, Dr. Muhammad Imran (University of Glasgow)
Project: "A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems"

- Co-led the literature review and manuscript writing, covering 114 references under faculty supervision.

- Classified 106 papers by SON function and publication period, identifying key research trends.
- Identified open problems, including conflicts between SON functions and SON's role in 6G.

Research Assistant, **Institute for Research in Fundamental Sciences (IPM)**
& **Sharif University of Technology**, Tehran, Iran Jun 2025–present
Supervisor: Dr. Arash Vaezi
Project: "Operational Resilience of Blockchain Remote Procedure Call Services"

- Co-developed a five-dimension capability model for analyzing blockchain RPC exposure.
- Analyzed legacy Geth remote-API attacks and the shift to Clef external signing.
- Mapped defenses across node, signer and provider layers and derived eight open problems.

Research Intern, **Politecnico di Milano**, Milan, Italy (Remote) May 2024–Aug 2024
Supervisor: Dr. Gianni Antichi

- Researched WebAssembly for systems acceleration and designed a benchmarking plan for CPU, memory, latency and throughput.

Research Intern, **Sharif University of Technology**, Tehran, Iran Oct 2023–Jan 2024
Supervisor: Dr. Bardia Safaei

- Simulated a decentralized FANET in NS-3 using ant-colony-based clustered routing.
- Implemented beaconing and neighbor discovery under UAV mobility, and analyzed energy–delivery trade-offs.

SELECTED TEACHING EXPERIENCE

Teaching Assistant, Sharif University of Technology

- **Computer Networks** — Dr. Mahdi Dolati Fall 2026, Winter 2026
- **Operating Systems** — Dr. Hossein Asadi Fall 2026
- **Operating Systems** — Dr. Hossein Asadi & Dr. Rasul Jalili Fall 2024
- **Computer Networks** — Dr. Hossein Peyvandi Winter 2024
- **Operating Systems** — Dr. Arash Vaezi [Piazza] Winter 2024
- **Computer Networks** — Dr. Kambiz Mizanian Fall 2023
- **Operating Systems** — Dr. Mohaddeseh Mirbeygi & Dr. MohammadAli Mirzaei Fall 2023

Designed and graded weekly assignments (120+ students in the largest classes), wrote rubrics, held office hours, guided projects and built autograders.

NOTABLE PROJECTS

SliceDB — Distributed Key-Value Store [GitHub Link]

- Co-developed an in-memory Go store with hash partitioning, leader–follower WAL replication and snapshot recovery.
- Implemented coordination with etcd leases, controller election and fenced metadata updates blocking stale controllers.
- Added online resharding, replica rebalancing, automatic failover and graceful leader handoff.

NYC Taxi Analytics on Highly Available HDFS [GitHub Link]

- Co-built a Dockerized HDFS cluster with replicated storage.
- Configured ZooKeeper-based NameNode failover and a logical HDFS nameservice for failover-aware client access.
- Analyzed NYC taxi trips using Java Spark and stored results in HDFS.

Distributed Log Analytics [GitHub Link]

- Co-developed Go services for Kafka ingestion and ClickHouse queries.
- Integrated Cassandra metadata, CockroachDB auditing and retention.

Product Retrieval and Shopping Agent [GitHub Link]

- Indexed 1M+ products for text search and 99K+ images using FAISS.
- Built a FastAPI shopping agent with memory and monitoring.

SDN Routing with Floodlight [GitHub Link]

- Developed a Java Floodlight controller module exposing Dijkstra-based shortest-path queries through a REST API.
- Designed an OpenFlow 1.3 testbed in Mininet with controlled link bandwidth and latency for routing experiments..

HONORS & AWARDS

Top 0.1% among 155,250 candidates, Iranian University Entrance Exam (B.Sc.) 2020
Two-Time National Hovercraft Competition Winner (High School) — Design & Control 2016–2017

SKILLS

Programming Languages: Java, Python, C/C++, Go, JavaScript, Bash

Frameworks & Libraries: PyTorch, TensorFlow, FAISS, NumPy, Matplotlib, scikit-learn

Tools: Git, Docker, NS-3, LaTeX

Web Programming: Vue, Flask, HTML5, CSS

Languages: Persian (Native), English (C1)